

High Slew Rate Micro-Power CMOS OTA with Class AB Input stage

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ABSTRACT

The design of a new single stage class AB operational amplifier is presented. Amplifier incorporates an input differential pair with minimum current selector and common-mode feedback to control the tail current. It shows small-signal behavior of a simple OTA, while class AB feedback provides high slew rate when large input signals are applied. Prototype circuit was manufactured in 2 μ m analog CMOS process. In the quiescent mode it consumes less than 20 μ A of supply current. Circuit has DC gain of 70dB, slew rate of 2.5 V/ μ s at 10pF load and 0.9 V/ μ s at 100pF load. THD is better than 90dB at 0.2 V_{P-P} input signal and CMRR is 110dB.

1. INTRODUCTION

One of the difficulties imposed by low power design constraints in an operational amplifier [1] is a limited slew rate. Requirements of low current consumption and frequency compensation in a traditional differential pair input stage inherently put a nonlinear limitation on the maximum rate of change of the output voltage when a large differential signal is applied to the input, although the signal itself can be well below the unity gain frequency of the amplifier [3].

For a typical input stage with a differential pair and current I_t , the maximum rate of the signal change at the output, so called slew rate, is limited by the speed of charging of load or compensation capacitance C_C . Slew rate is given by:

$$SR = \frac{dV_{out}}{dt} = \frac{I_t}{C_C} \quad (1)$$

If we substitute the small-signal equation $C_C = g_m/\omega_t$ in (1):

$$SR = \frac{I_t \omega_t}{g_m} = \frac{I_t \omega_t}{\sqrt{\mu C_{ox} \left(\frac{W}{L}\right)_1 I_t}} = V_{Dsat_{M1}} \omega_t \quad (2)$$

From (2) one can see that the only ways to increase the slew rate in this structure are to increase either V_{Dsat} , or ω_t , or both [2, p. 228]. But ω_t is controlled by other factors, like power consumption and the bandwidth of

the second stage. V_{Dsat} can be increased by larger I_t (and power consumption) or lower W/L , latter leading to the lower transconductance g_m and lower gain of the first stage and increased noise and input offset of the amplifier. Although these ways are often employed to achieve higher SR, they prove to be inefficient and structurally unsound.

Naturally, the way to increase a slew rate of the amplifier would be to change the internal structure of a traditional voltage-feedback op amp to remove the tail current limitation. There have been a number of circuit topologies suggested in literature [3, 4, 5, 6, 7, 8, 9, 10, 11]. For the lack of space for discussion the reader is referred to these references and [12] for details.

The approach chosen for this work would be to introduce a common-mode feedback loop and keep the basic differential pair structure with its benefits of excellent small signal performance. The feedback would increase the tail current when the amplifier goes to the slew mode. provided a very good slew rate performance without degrading other parameters, such as noise, linearity and offset.

2. CLASS AB INPUT STAGE WITH COMMON-MODE FEEDBACK

Differential circuits with common-mode feedback can be represented by a signal flow graph [13, 14]. Nodes of the graph are corresponding to voltages or currents in the circuit and branches are voltage/currents gain or transconductance/impedance devices. In fact, a conventional differential pair of transistors can be looked at as a system with linear class A common-mode feedback. The reader can refer to [13] for more information and mathematical proof of the equivalence of the signal graph to transistor and amplifier circuits.

The proposed class AB structure that eliminates tail current limitation is shown in Figure 1. The similar structure was recently independently reported by R. Wassenaar in [15, p. 301]. Class AB operation is achieved by the addition of the non-linear block performing a sort of voting operation selecting minimum current of I_1 , I_2 . On the signal graph, this can be shown

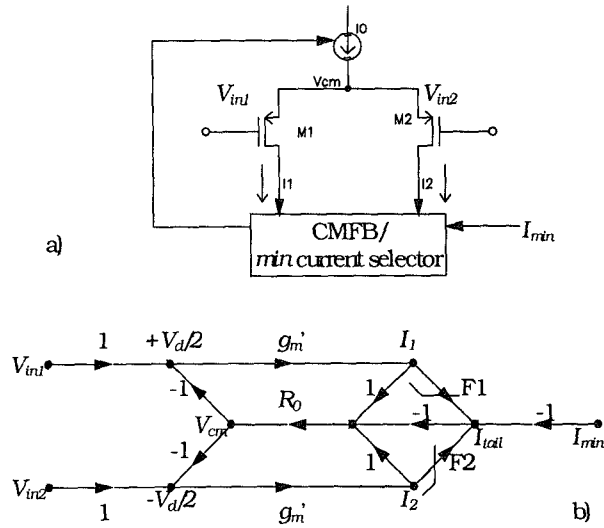


Figure 1: Proposed structure and signal graph of a differential pair with class AB common-mode feedback.

as branches with nonlinear transfer functions $y=F(x)$, such that output has a lower bound and increases whenever input is decreasing.

The output signal of *min* current selector block controls the tail current source of the differential pair.

The structure operates as follows: when either of I_1 or I_2 is decreasing and $M1$ or $M2$ starts shutting off, the output signal from CMFB block increases the tail current of the differential pair. In the quiescent mode or in presence of the small signal at the input, the tail current of the stage is defined by I_{min} , which is subtracted from the sum of I_1+I_2 .

A CMOS transistor implementation of the class AB input pair is shown in Figure 2. It can be easily seen from the circuit diagram that this circuit retains the basic input differential pair $M1, M2$ and its tail current source $M3$. All other devices implement the common-mode feedback which provides class AB behavior of $M3$. The circuit operates as follows:

The drain currents of transistors $M1a, M2a$ are proportional to the currents of the main pair $M1, M2$ with the ratio $1/A$, defined by the ratio of W/L of corresponding devices. Transistors $M8, M34$ are fixed current sources with drain currents $I_{D M8} = I_{D M34} = I_3$ defined by the current mirror $M7$. Devices $M6, M16$ have both of their drains connected to current source I_9 . This high impedance node controls the gate of $M3$, which closes the feedback loop. By connecting drains of $M6, M16$ together and biasing them by a constant voltage V_b we can perform a minimum-like voting function on voltages V_1, V_2 at nodes 1 and 2.

In the quiescent point, without any differential input voltage applied to the gates of $M1, M2$, voltage $V_1=V_2$; current I_9 splits evenly between $M6$ and $M16$ and flows into $M8, M34$. The rest of the current in $M8, M34$ is

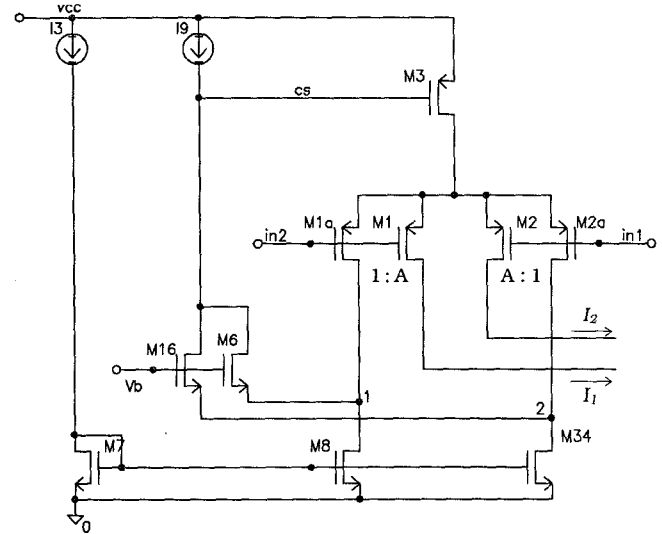


Figure 2: Input differential pair with proposed class AB feedback circuitry.

supplied by input devices $M1a$ and $M2a$ correspondingly. Therefore, the quiescent biasing point of input pair can be expressed as:

$$I_1 = I_2 = A \left(I_3 - \frac{I_9}{2} \right) \quad (3)$$

Operation of the stage is illustrated by the DC sweep simulation results in Figure 3. In the top plot are the curves of the output currents of the main pair $M1, M2$ versus differential input voltage V_{in} .

When a small differential voltage is applied to input nodes of the stage, difference in currents of $M1a, M2a$ is compensated by the difference in currents of $M6$ and $M16$, maintaining sums:

$$I_{M1a} + I_{M16} = I_{M2a} + I_{M6} = I_3; \quad I_{M16} + I_{M6} = I_9 \quad (4)$$

This resulting in no net change in the gate voltage of $M3$. Therefore, the stage behaves as a conventional differential pair, which can be seen from the bottom plot in Figure 3.

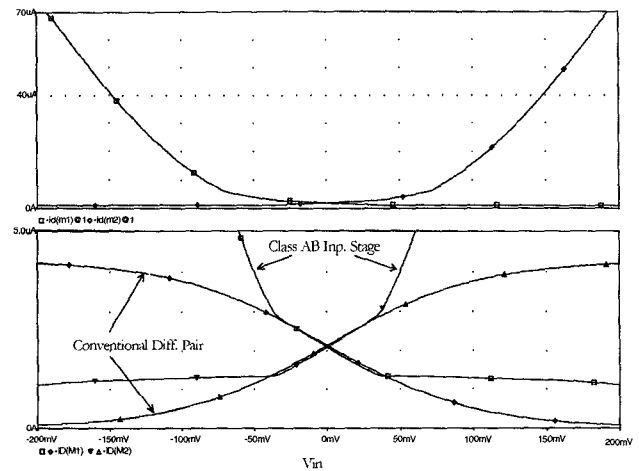


Figure 3: DC simulation results of class AB input pair.

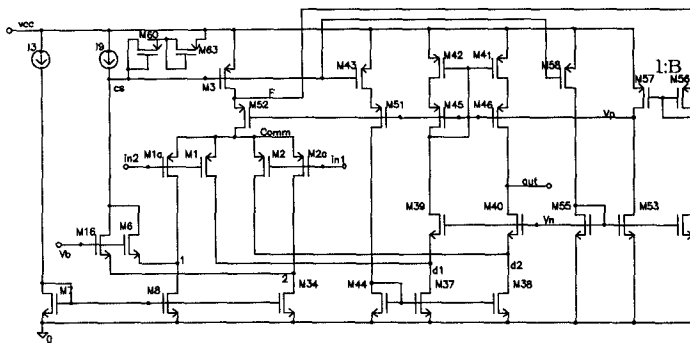


Figure 4: Full circuit diagram of the class AB single stage operational amplifier.

If input voltage is increased further, one of the transistors $M6$, $M16$ goes into shut off mode (since I_{M1a} or $I_{m2a} > I_3$), while the other one engages in the control loop. If, for example, the I_{M2a} is increased and I_{M1a} is reduced, the voltage at node 2 goes up and shuts off $M16$, while the V_I goes down, thus bringing down the voltage at the gate of $M3$. This increases the tail current of the stage, maintaining the minimum current in $M1a$ and $M1$ and increasing the current in $M2$. The minimum current of $M1$ is defined by:

$$I_{D_{M1} \min} = A(I_3 - I_9) \quad (5)$$

As it is shown in the top plot of Figure 3, output currents of the stage have a typical class AB behavior. In the bottom plot are the same currents with Y axis zoomed in around the quiescent point. A slight decrease in the drain current of MI in the right part of the plot when the input voltage is further increased can be explained by the finite loop gain.

Also included in the bottom plot are output currents of a conventional differential pair for comparison of small and large signal performance.

A big advantage of the proposed class AB input stage is that it has a conventional linear g_m behavior around zero input voltage, but unlike the conventional input pair, which has a limited current output, the proposed stage increases its output current if the differential voltage is increased.

Since the input stage is usually loaded on a capacitor, this characteristic provides a big increase in the slew rate without any non-linearity in the small signal behavior and causes no stability problems. Unlike previously proposed class AB stages with similar structure, the maximum tail current in this circuit does not have a non-linear limitation, which lets it work on much larger capacitive loads. In fact, as will be shown later, we should install clamping devices on the gate of $M3$ to prevent excessive sinking of the current in those cases.

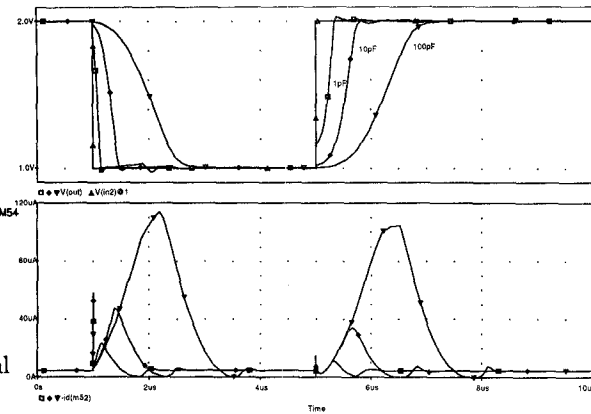


Figure 5: Unity gain step response and tail current during transient.

3. CIRCUIT OF THE CLASS AB SINGLE STAGE OP AMP

The circuit diagram of the folded cascode amplifier incorporating the proposed class AB input is shown in Figure 4. The circuit was simulated using SPICE Level 2 models for MOSIS 2 μ m analog CMOS process. The amplifier is biased by currents $I_3=0.6\mu\text{A}$, $I_2=0.3\mu\text{A}$, ratio $A=K_{M1}/K_{M1a}=5$. From equation (3) we can obtain quiescent currents of the input transistors $I_1=I_2=2\mu\text{A}$. DC simulation results were discussed above.

The response of the proposed amplifier to 1V positive and negative input steps is shown in

Figure 5. The top plot shows input step voltage and output response under different capacitive loads 1pF, 10pF, 100pF. The amplifier is stable under all conditions. The bottom plot illustrates changes of the tail current of the input pair over transient process.

The graphs show that the amplifier has a non-linear dependence of the settling time versus load capacitance. It can be explained by the finite bandwidth of class AB feedback loop in the input stage.

The comparison of step responses of the proposed amplifier and the conventional one is shown in Figure 6.

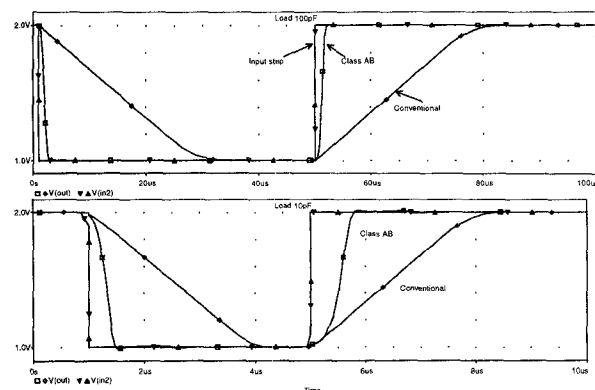


Figure 6: Comparison of slew rates of the conventional and class AB amplifier. Note different time scales.

The top plot shows output with 100pF load and the bottom is for 10pF.

From this data it can be seen that the proposed technique is more advantageous when the amplifier is loaded on a large capacitor, although it provides a very significant improvement even with small capacitive loads.

The comparison data of dynamic performance of the proposed amplifier and a conventional one are shown in Table 1. This data and plots above show that the novel amplifier provides a large signal settling time that is equal or better than the settling time of an ideal op amp for capacitive loads more than 10pF.

4. CONCLUSION

A new structure and a topology of class AB input stage has been introduced. The circuit has very good power efficiency and small-signal characteristics of a conventional differential pair while having high output current when large input voltage is applied. A new single stage amplifier circuit based on this topology was proposed. The amplifier works from low supply voltage, has a superior slew rate and good small-signal performance with minimal quiescent power consumption. This makes this circuit very desirable for applications that require large signal handling with low distortion, for instance switched capacitor circuits, audio and video signal processing in portable equipment.

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Load	Parameter	Novel Class AB amplifier	Conventional op amp
1pF	f_t	4.2 MHz	
	Slew rate *	3.5...7.0 V/ μ s**	2.2 V/ μ s
	1% Settling *	1.1 μ s	0.6...0.85 μ s
10pF	f_t	640 kHz	
	Slew rate *	2.5...2.9 V/ μ s**	0.34 V/ μ s
	1% Settling *	1.4...1.8 μ s**	3.1 μ s
100pF	f_t	67 kHz	
	Slew rate *	0.9 V/ μ s	0.036 V/ μ s
	1% Settling *	2.0 μ s	31.2 μ s

* Slew rate and settling time are given for 1V input voltage step in unity buffer feedback configuration.

** Range of values is shown for positive and negative steps.

Table 1: Comparison of dynamic performance.